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COURSE CODE: SBT-DF204
LAB NAME: INVESTIGATING RHION POSSESSION EVIDENCE
DATE: 26/10/2026
INSTRUCTOR: AMINU IDRIS

This individual case study brings together disk-image recovery, steganography, FTP, HTTP, and Wireshark analysis. You will work from the authorised course evidence package and produce an evidence-led account of what the artefacts do and do not establish. Your report must clearly separate direct observations from inferences about file transfer, concealment, possession, or the person who used a device.

Case scenario

This is a fictional training investigation concerning imagery that may meet an illegal-possession threshold in the case scenario. The evidence package includes a seized USB disk image and network traces associated with FTP and HTTP activity. Investigators need to know whether deleted files, hidden data, reconstructed network objects, and the timing of recorded transfers form a defensible sequence. The task is an offline examination of supplied training artefacts; it does not authorise any contact with people, accounts, systems, hosts, or services named in the material.

Case materials

Use only the materials supplied through E-Campus and the accompanying teaching decks. The course sequence includes a USB DD image, an FTP trace (rhino.log), HTTP traces (rhino2.log and rhino3.log), and supporting recovery and Wireshark guidance. Use the exact filename in your evidence package if it differs from the deck. Before analysis, create a working copy and record integrity information for every original file. Do not alter, shred, or dispose of original course material before your work has been assessed.

What you must submit

Item	Required output	What to provide
1	Forensic working set	Record the supplied disk image, packet captures, and any recovered artefact by original file name, byte size, SHA-256 value, source location, and working-copy location. Preserve the original course material unchanged.
2	Recovery and analysis record	Keep labelled output or screenshots for the disk-image layout, file recovery, metadata review, steganography testing, FTP and HTTP analysis, reconstructed objects, and validation checks.
3	Correlated timeline	Create a UTC, time-ordered account with at least five material events across at least three artefact families. Cite packet numbers or streams, file paths, hashes, timestamps, and evidence-log IDs.
4	Student report	Submit one report in PDF or DOCX format with a completed evidence log and timeline appendix. Include only the supporting files requested by your lecturer.

Investigation task

Conduct the examination in stages: preserve the supplied material; recover and validate relevant files from the disk image; examine media for hidden content; reconstruct and validate material FTP and HTTP objects; then correlate the artefacts in a single UTC timeline. Base every finding on identifiable records such as a source filename, hash, partition or metadata record, packet number, TCP stream, timestamp, response header, tool output, or screenshot. The questions below define the minimum scope of the report.

No	Investigation question	Minimum evidence to discuss
1	How did you preserve the case materials?	For each supplied disk image and capture, record source, original file name, byte size, SHA-256 value, working-copy method, tool version, and file-access precautions.
2	What does the USB disk image show before recovery?	Document the partition layout, file-system information, allocated or unallocated area examined, relevant directory or metadata records, and any limits of file-system metadata.
3	What deleted or recoverable files are supported?	Describe the recovery method, output folder, recovered-file type, hash, visual or metadata check, and whether files are unique or duplicates. Do not report deck answers as your own findings.
4	Is there evidence of hidden content in the recovered media?	Record metadata review, steganography detection results, the authorised offline recovery method used, recovered output, hashes, and the evidence that supports or weakens a hiding interpretation.
5	What does the FTP capture establish?	Identify relevant sessions, transfer direction, packet or stream references, server response, object boundaries, reconstructed file validation, and timestamps. Treat any visible credential as sensitive evidence; do not reuse it.
6	What does the protected archive analysis establish?	Show that the reconstructed archive is complete, identify the evidence of encryption, and document any authorised course-only password-recovery workflow. Compare recovered content with other evidence using hashes where available.
7	What does the HTTP capture establish?	Identify material requests and responses, client and server endpoints, content type, stream or packet references, object extraction and validation. Do not execute any reconstructed executable or contact a recorded host.
8	What conclusion can you defend?	Build a UTC timeline, distinguish direct observations from inference, state a confidence level, and explain at least three limitations or alternative explanations concerning possession, transfer, and user attribution.

Technical starting points

The following are course-aligned starting points, not a substitute for documenting your own method. Work only on copies. Replace placeholders with values from your evidence and preserve the outputs that substantiate each decision.

Purpose	Course-aligned starting point

Preserve and identify evidence	Create read-only originals and working copies. Record SHA-256 values before and after copying; use file type and hex inspection only on working copies.
Examine the disk image	Use the Sleuth Kit workflow demonstrated in class to review media layout and file-system records (for example, mmls, fsstat, fls, and icat where appropriate).
Recover deleted files	Use the authorised PhotoRec or file-carving workflow on the working copy. Record the recovery scope selected and the output location; hash every material recovered file.
Review media and hidden data	Review file type and metadata, then use the supplied steganography workflow to detect and, if justified, recover embedded course content. Keep detector output and recovery notes.
Review FTP traffic	Use Wireshark display filtering and Follow TCP Stream to locate control and data sessions. Preserve packet or stream numbers, direction, timestamps, and raw object boundaries.
Review HTTP traffic	Use http, http.request, http.response, and tcp.stream eq <number> as starting filters. Confirm reassembly, response headers, content type, and file boundaries before exporting a working copy.
Validate reconstructed objects	Use magic number and trailer checks, file-type output, byte size, and SHA-256 hashes. A matching hash can establish binary equality but does not identify a person or original source by itself.

Evidence interpretation safeguards

- Use a hash and read-only preservation record for each original. A recovered working copy or carved file must remain traceable to its source and recovery method.
- For network objects, preserve the relevant packet range or TCP-stream reference, transfer direction, and reassembly method. A single packet is not necessarily a complete file.
- Visible FTP or email credentials are sensitive evidence. Record only what is necessary for assessment, redact them in screenshots where appropriate, and never reuse them outside the supplied course artefacts.
- An authorised password-recovery step may be used only against the supplied training archive and only within the course lab. Do not apply recovered passwords, wordlists, or methods to any other file, account, host, or service.
- A file hash, IP address, host name, username, or directory record can corroborate a relationship, but it does not alone prove original source, completion of an action, physical possession, or human identity.
- Do not execute a reconstructed executable, connect to a recorded endpoint, submit material to public scanners, or upload case evidence outside the approved lab. Follow lecturer-approved secure-disposal instructions only after assessment.

Required report structure

1. Case reference and student identification. Include your full name, registration number, course code, and case title.
2. Executive conclusion. State a concise, cautious conclusion and confidence level. Do not present a person-level attribution unless the artefacts support it.
3. Evidence acquisition and integrity. Record sources, filenames, sizes, hashes, read-only originals, working copies, tools, versions, and access controls.
4. Method. Explain the disk-image, recovery, steganography, FTP, HTTP, Wireshark, object-validation, and time-normalisation methods used.
5. Findings. Answer every investigation question with a precise evidence reference: file path, hash, metadata record, packet number, stream, timestamp, header, or screenshot ID.
6. Timeline and correlation. Present the material sequence in UTC. State how artefacts corroborate, conflict with, or fail to establish each proposed relationship.

7. Conclusion and limitations. Explain gaps, alternative explanations, duplicate files, partial captures, encryption, timestamp uncertainty, and the distinction between technical account evidence and a person at a keyboard.
8. Appendices. Include labelled screenshots, selected tool output, the completed timeline, the completed evidence log, and only the supporting material requested by your lecturer.

Timeline template

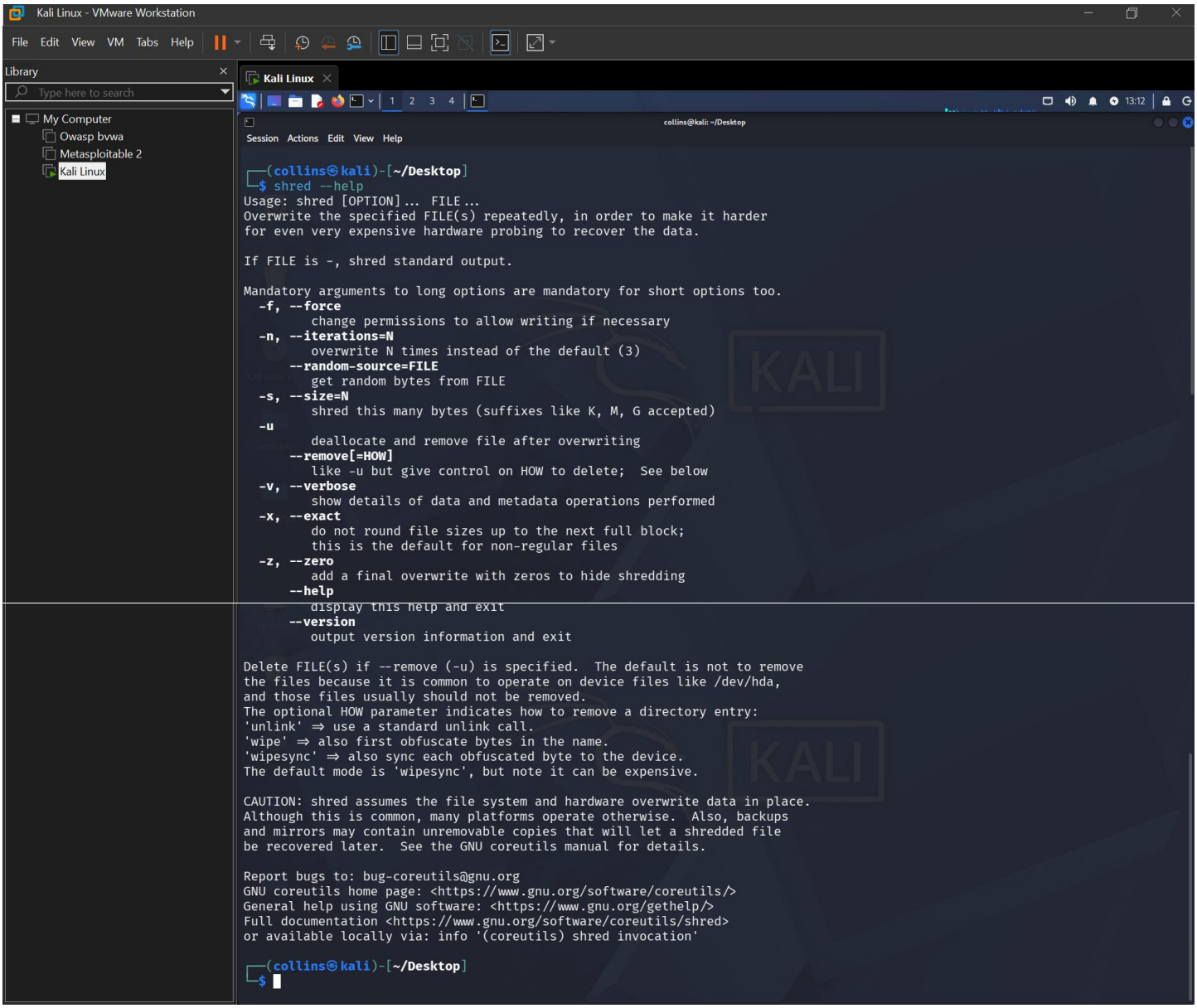
Complete a version of this table in your report or evidence appendix. Add rows for all material events and keep the main timeline in UTC.

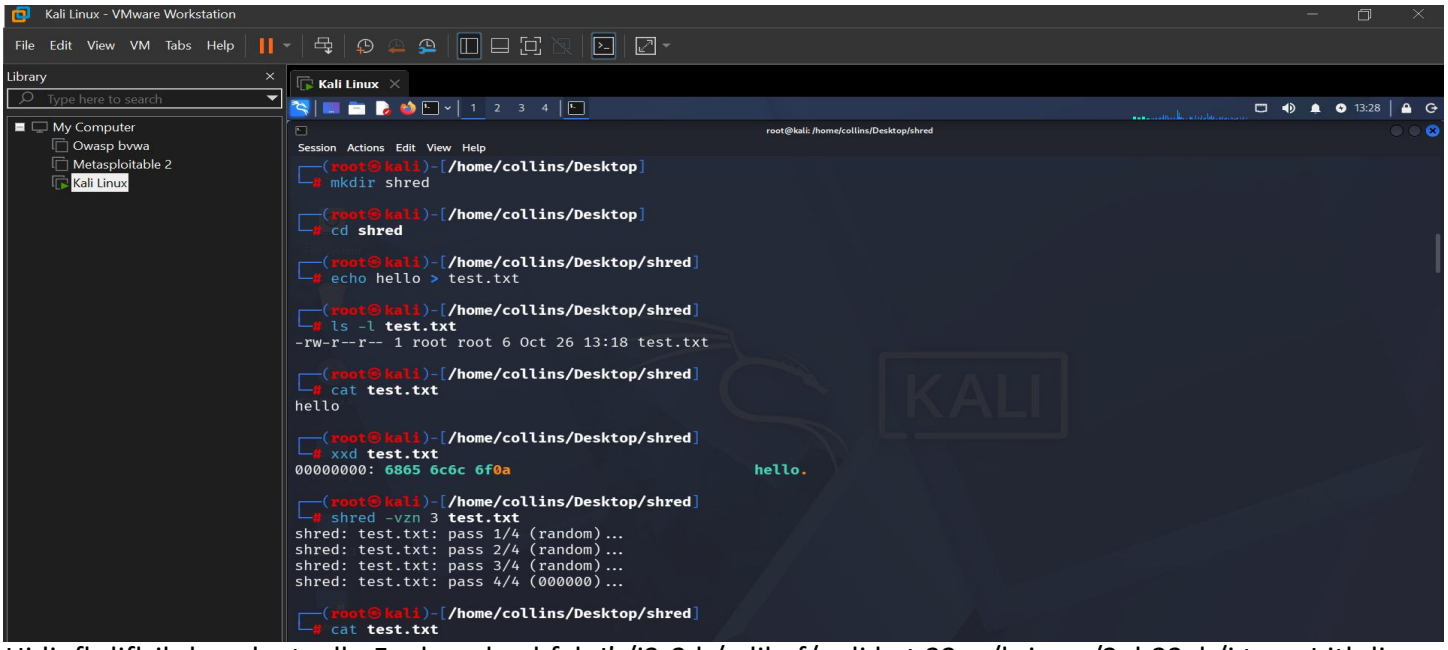
Stage	UTC timestamp	Evidence ID	Source artefact	Observed fact	Interpretation and limit
Media preservation					
Disk image / file-system review					
Recovered or hidden content					
FTP transfer activity					
HTTP transfer activity					
Corroborated conclusion					

Evidence log template

Use an evidence log that lets an assessor locate the exact source file, tool, command or workflow, packet or stream, record, hash, and screenshot that supports each finding.

Evidence ID	Tool and method	Source record, stream, or path	Observed fact	Screenshot or appendix reference
E01				
E02				
E03				
E04				
E05				
E06				
E07				





Hjdiofksljflsjhdvo;ebwtoglh. Fnebg;o.luwhfeb;l';/i3r2.b/u.ljhgf/uelj.hgt;23oy/l.rjg;yo/2r.h23gl;/j.tonp l;jth;ljpue

Metasploitable 2
Kali Linux

```
┌──(root㉿kali)-[/home/collins/Desktop/shred]
└─# xxd test.txt
00000000: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000030: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000040: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000050: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000060: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000070: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000140: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000150: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000160: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000170: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000180: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000190: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000001b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................

┌──(root㉿kali)-[/home/collins/Desktop/shred]
└─# xxd test.txt | tail
00000f60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000f70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000f80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000f90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000fa0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000fb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000fc0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000fd0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000fe0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000ff0: 0000 0000 0000 0000 0000 0000 0000 0000  ................

┌──(root㉿kali)-[/home/collins/Desktop/shred]
└─# ls -l test.txt
-rw-r--r-- 1 root root 4096 Oct 26 13:25 test.txt
```

Metasploitable 2
Kali Linux

```
┌──(root㉿kali)-[/home/collins/Desktop/shred]
└─# wget -q https://www.dropbox.com/s/p575df3fui42d88/660×450-city-shot.jpg

┌──(root㉿kali)-[/home/collins/Desktop/shred]
└─# ls -l 660×450-city-shot.jpg
-rw-r--r-- 1 root root 434176 Oct 26 13:38 660×450-city-shot.jpg

┌──(root㉿kali)-[/home/collins/Desktop/shred]
└─# xxd 660×450-city-shot.jpg
00000000: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000010: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000020: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000030: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000040: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000050: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000060: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000070: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000080: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000090: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000a0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000b0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000c0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000d0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000e0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
000000f0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000100: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000110: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000120: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00000130: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069e70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069e80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069e90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069ea0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069eb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069ec0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069ed0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069ee0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069ef0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f00: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f10: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f20: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f30: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f40: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f50: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f60: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f70: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f80: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069f90: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069fa0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069fb0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069fc0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069fd0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069fe0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
00069ff0: 0000 0000 0000 0000 0000 0000 0000 0000  ................
```

Jhblqw .\guqlwb\efhgskjuhikltfasugkl;pwoyitys;kjhjfpiwtpi;hwnkulip’w;iulyktrewsfag;s’poiluyjtrdanm,jk’lloukjhl

Kali Linux

```
root@kali:~/rhion# wget -q https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Illegal_Possession_Images/USB_image/DFRWS2005-RODEO.zip

root@kali:~/rhion# unzip DFRWS2005-RODEO.zip
Archive:  DFRWS2005-RODEO.zip
  inflating: RHINOUSB.dd
  inflating: rhino.log
  inflating: rhino2.log
  inflating: rhino3.log

root@kali:~/rhion#

root@kali:~/rhion# openssl dgst -md5 rhino.log
MD5(rhino.log)= c0d0093eb1664cd7b73f3a5225ae3f30

root@kali:~/rhion# openssl dgst -md5 rhino2.log
MD5(rhino2.log)= cd21eaf4acfb50f71ffff857d7968341

root@kali:~/rhion# openssl dgst -md5 rhino3.log
MD5(rhino3.log)= 7e29f9d67346df25faaf18efcd95fc30

root@kali:~/rhion# openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54
```

```
root@kali:~/rhion# fdisk -l RHINOUSB.dd
Disk RHINOUSB.dd: 247.48 MiB, 259506176 bytes, 506848 sectors
Units: sectors of 1 * 512 = 512 bytes
Sector size (logical/physical): 512 bytes / 512 bytes
I/O size (minimum/optimal): 512 bytes / 512 bytes
Disklabel type: dos
Disk identifier: 0x00000000

root@kali:~/rhion# fsstat -b 512 -o 0 RHINOUSB.dd
FILE SYSTEM INFORMATION
--------------------------------------------
File System Type: FAT16

OEM Name: mkdosfs
Volume ID: 0x4092d9d1
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: FAT16

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 506847
* Reserved: 0 - 0
** Boot Sector: 0
* FAT 0: 1 - 248
* FAT 1: 249 - 496
* Data Area: 497 - 506847
** Root Directory: 497 - 528
** Cluster Area: 529 - 506840
** Non-clustered: 506841 - 506847

METADATA INFORMATION
--------------------------------------------
Range: 2 - 8101622
Root Directory: 2
```

```
CONTENT INFORMATION
--------------------------------------------
Sector Size: 512
Cluster Size: 4096
Total Cluster Range: 2 - 63290

FAT CONTENTS (in sectors)
--------------------------------------------
529-536 (8) -> EOF
537-544 (8) -> EOF

root@kali:~/rhion# fls -o 0 RHINOUSB.dd
r/r 4:  gumbo1.txt
r/r 6:  gumbo2.txt
v/v 8101619:    $MBR
v/v 8101620:    $FAT1
v/v 8101621:    $FAT2
V/V 8101622:    $OrphanFiles
```

Owasp bvwa
Metasploitable 2
Kali Linux

Session Actions Edit View Help

```
root@kali:~/rhion# fls
Missing image name
usage: fls [-adDFlhpruvV] [-f fstype] [-i imgtype] [-b dev_sector_size] [-m dir/] [-o imgoffset] [-z ZONE] [-s seconds] image
[images] [inode]
        If [inode] is not given, the root directory is used
        -a: Display "." and ".." entries
        -d: Display deleted entries only
        -D: Display only directories
        -F: Display only files
        -l: Display long version (like ls -l)
        -i imgtype: Format of image file (use '-i list' for supported types)
        -b dev_sector_size: The size (in bytes) of the device sectors
        -f fstype: File system type (use '-f list' for supported types)
        -m: Display output in mactime input format with
              dir/ as the actual mount point of the image
        -h: Include MD5 checksum hash in mactime output
        -o imgoffset: Offset into image file (in sectors)
        -P pooltype: Pool container type (use '-P list' for supported types)
        -B pool_volume_block: Starting block (for pool volumes only)
        -S snap_id: Snapshot ID (for APFS only)
        -p: Display full path for each file
        -r: Recurse on directory entries
        -u: Display undeleted entries only
        -v: verbose output to stderr
        -V: Print version
        -z: Time zone of original machine (i.e. EST5EDT or GMT) (only useful with -l)
        -s seconds: Time skew of original machine (in seconds) (only useful with -l & -m)
        -k password: Decryption password for encrypted volumes

root@kali:~/rhion#
```

Metasploitable 2
Kali Linux

```
root@kali:~/rhion# icat -o 0 RHINOUSB.dd 4 > gumbo1_output.txt

root@kali:~/rhion# icat -o 0 RHINOUSB.dd 6 > gumbo2_output.txt

root@kali:~/rhion# ls -l *.txt
-rw-r--r-- 1 root root 2815 Oct 26 14:39 gumbo1_output.txt
-rw-r--r-- 1 root root 1293 Oct 26 14:39 gumbo2_output.txt

root@kali:~/rhion# cat gumbo1_output.txt | head -n 5
SHRIMP AND TASSO GUMBO

Associate Food Editor: Alexis Touchet
Father: Rodney Miller, Abbeville, LA

root@kali:~/rhion# cat gumbo2_output.txt | head -n 5
SHRIMP AND ANDOUILLE SAUSAGE GUMBO

1/2 cup vegetable oil
1/2 cup all purpose flour
4 celery stalks, coarsely chopped

root@kali:~/rhion#

root@kali:~/rhion# openssl dgst -md5 RHINOUSB.dd
MD5(RHINOUSB.dd)= 80348c58eec4c328ef1f7709adc56a54

root@kali:~/rhion# dd if=RHINOUSB.dd of=RHINOUSB_My_COPY.dd
506848+0 records in
```

Metasploitable 2
Kali Linux

```
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

  PhotoRec is free software, and
comes with ABSOLUTELY NO WARRANTY.

Select a media and choose 'Proceed' using arrow keys:
>Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)




>[Proceed ]  [  Quit  ]

Note:
Disk capacity must be correctly detected for a successful recovery.
If a disk listed above has an incorrect size, check HD jumper settings and BIOS
detection, and install the latest OS patches and disk drivers.
```


Owasp bvwa
Metasploitable 2
Kali Linux
Session Actions Edit View Help
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)
Partition Start End Size in sectors
Unknown 0 0 1 1021 6 60 506848 [Whole disk]
> P FAT16 0 0 1 1021 6 60 506848
>[Search] [Options] [File Opt] [Quit]
Start file recovery
Session Actions Edit View Help
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
P FAT16 0 0 1 1021 6 60 506848
To recover lost files, PhotoRec needs to know the filesystem type where the
file were stored:
[ext2/ext3] ext2/ext3/ext4 filesystem
>[Other] FAT/NTFS/HFS+/ReiserFS/...
Session Actions Edit View Help
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org
P FAT16 0 0 1 1021 6 60 506848
Please choose if all space needs to be analysed:
>[Free] Scan for files from FAT16 unallocated space only
[Whole] Extract files from whole partition
PhotoRec 7.2, Data Recovery Utility, February 2024
Please select a destination to save the recovered files to.
Do not choose to write the files to the same partition they were stored on.
Keys: Arrow keys to select another directory
C when the destination is correct
Q to quit
Directory /root/rhion
>drwxr-xr-x 0 0 4096 26-Oct-2026 14:47 .
drwx------ 0 0 4096 26-Oct-2026 13:46 ..
-rw-r--r-- 0 0 3614418 26-Oct-2026 14:08 DFRWS2005-RODEO.zip
-rw-r--r-- 0 0 259506176 1-May-2004 02:29 RHINOUSB.dd
-rw-r--r-- 0 0 259506176 26-Oct-2026 14:45 RHINOUSB_My_COPY.dd
-rw-r--r-- 0 0 2815 26-Oct-2026 14:39 gumbo1_output.txt
-rw-r--r-- 0 0 1293 26-Oct-2026 14:39 gumbo2_output.txt
-rw-r--r-- 0 0 40960 26-Oct-2026 14:47 photorec.se2
-rw-r--r-- 0 0 3187907 27-Apr-2004 00:42 rhino.log
-rw-r--r-- 0 0 292604 28-Apr-2004 23:50 rhino2.log
-rw-r--r-- 0 0 226094 28-Apr-2004 23:50 rhino3.log

Owasp bvwa
Metasploitable 2
Kali Linux

```
Session  Actions  Edit  View  Help
PhotoRec 7.2, Data Recovery Utility, February 2024
Christophe GRENIER <grenier@cgsecurity.org>
https://www.cgsecurity.org

Disk RHINOUSB_My_COPY.dd - 259 MB / 247 MiB (RO)
     Partition                  Start        End    Size in sectors
   P FAT16                    0   0  1  1021   6 60     506848


132 files saved in /root/rhion/recup_dir directory.
Recovery completed.

You are welcome to donate to support and encourage further development
https://www.cgsecurity.org/wiki/Donation

[ Quit ]
```

```
root@kali:~/rhion# ls -l
total 514068
-rw-r--r-- 1 root root   3614418 Oct 26 14:08 DFRWS2005-RODEO.zip
-rw-r--r-- 1 root root      2815 Oct 26 14:39 gumbo1_output.txt
-rw-r--r-- 1 root root      1293 Oct 26 14:39 gumbo2_output.txt
-rw-r--r-- 1 root root     40960 Oct 26 14:47 photorec.se2
drwxr-xr-x 2 root root      4096 Oct 26 14:54 recup_dir.1
-rw-r--r-- 1 root root    292604 Apr 28  2004 rhino2.log
-rw-r--r-- 1 root root    226094 Apr 28  2004 rhino3.log
-rw-r--r-- 1 root root   3187907 Apr 27  2004 rhino.log
-rw-r--r-- 1 root root 259506176 May  1  2004 RHINOUSB.dd
-rw-r--r-- 1 root root 259506176 Oct 26 14:45 RHINOUSB_My_COPY.dd

root@kali:~/rhion# ls recup_dir.1
f0000545.txt                                  f0341081.txt  f0352905.txt  f0364729.txt  f0376553.txt  f0388377.txt
f0104057.jpg                                  f0341601.txt  f0353425.txt  f0365249.txt  f0377353.txt  f0388889.txt
f0104249.jpg                                  f0342113.txt  f0353937.txt  f0365761.txt  f0377585.txt  f0389409.txt
f0105065.jpg                                  f0342625.txt  f0354449.txt  f0366273.txt  f0378097.txt  f0389921.txt
f0105873.jpg                                  f0343137.txt  f0354961.txt  f0366785.txt  f0378609.txt  f0390433.txt
f0106393.jpg                                  f0343657.txt  f0355481.txt  f0367305.txt  f0379129.txt  f0390945.txt
f0106409.jpg                                  f0344585.txt  f0355993.txt  f0367817.txt  f0379641.txt  f0391465.txt
f0106865.gif                                  f0344681.txt  f0356505.txt  f0368329.txt  f0380153.txt  f0391977.txt
f0106889.gif                                  f0345201.txt  f0357017.txt  f0368841.txt  f0380665.txt  f0392489.txt
f0106905.txt                                  f0345713.txt  f0357537.txt  f0369361.txt  f0381185.txt  f0393001.txt
f0335017_She_died_in_February_at_the_age_of_74.doc  f0346225.txt  f0358049.txt  f0369873.txt  f0381697.txt  f0393737.txt
f0335081.jpg                                  f0346737.txt  f0358561.txt  f0370385.txt  f0382209.txt  f0394033.txt
f0335601.txt                                  f0347257.txt  f0359073.txt  f0370897.txt  f0382721.txt  f0394545.txt
f0335945.txt                                  f0347769.txt  f0359593.txt  f0371417.txt  f0383241.txt  f0395065.txt
f0336457.txt                                  f0348281.txt  f0360105.txt  f0371929.txt  f0383753.txt  f0395577.txt
f0336969.txt                                  f0348793.txt  f0360969.txt  f0372441.txt  f0384265.txt  f0396089.txt
f0337489.txt                                  f0349313.txt  f0361137.txt  f0372953.txt  f0384777.txt  f0396601.txt
f0338001.txt                                  f0349825.txt  f0361649.txt  f0373473.txt  f0385297.txt  report.xml
f0338513.txt                                  f0350337.txt  f0362161.txt  f0373985.txt  f0385809.txt
f0339025.txt                                  f0350849.txt  f0362673.txt  f0374497.txt  f0386321.txt
f0339545.txt                                  f0351369.txt  f0363193.txt  f0375009.txt  f0386833.txt
f0340057.txt                                  f0351881.txt  f0363705.txt  f0375529.txt  f0387353.txt
f0340569.txt                                  f0352393.txt  f0364217.txt  f0376041.txt  f0387865.txt

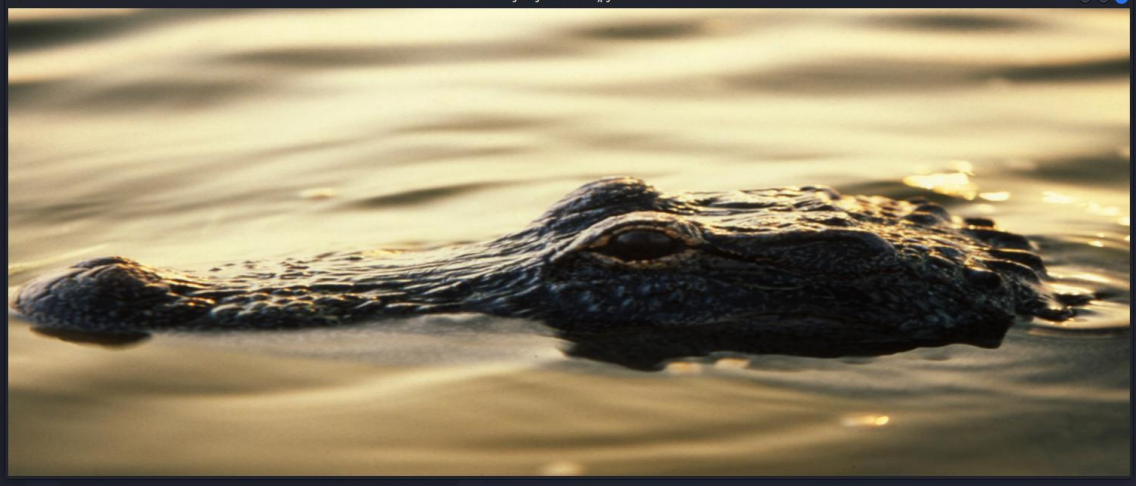
root@kali:~/rhion# cd recup_dir.1

root@kali:~/rhion/recup_dir.1# ls -l *.jpg
-rw-r--r-- 1 root root  95814 Oct 26 14:54 f0104057.jpg
-rw-r--r-- 1 root root 415534 Oct 26 14:54 f0104249.jpg
-rw-r--r-- 1 root root 411361 Oct 26 14:54 f0105065.jpg
-rw-r--r-- 1 root root 264600 Oct 26 14:54 f0105873.jpg
-rw-r--r-- 1 root root   6809 Oct 26 14:54 f0106393.jpg
-rw-r--r-- 1 root root 230665 Oct 26 14:54 f0106409.jpg
-rw-r--r-- 1 root root 264600 Oct 26 14:54 f0335081.jpg
```

Owasp bwa
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Session Actions Edit View Help

ImageMagick: f0104249.jpg



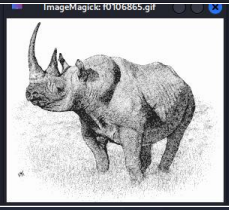
```
root@kali:~/rhion/recup_dir.1# display f0104249.jpg
```

```
root@kali:~/rhion/recup_dir.1# ls -l *.gif
-rw-r--r-- 1 root root 11407 Oct 26 14:54 f0106865.gif
-rw-r--r-- 1 root root  4105 Oct 26 14:54 f0106889.gif

root@kali:~/rhion/recup_dir.1# display f0106865.gif
^C

root@kali:~/rhion/recup_dir.1# display f0106865.gif
```

ImageMagick: f0106865.gif



```
root@kali:~/rhion/recup_dir.1# ls -l *.doc
-rw-r--r-- 1 root root 30720 Aug  9  2005 f0335017_She_died_in_February_at_the_age_of_74.doc

root@kali:~/rhion/recup_dir.1#
```

To return to your computer, move the mouse pointer outside or press Ctrl+Alt.

```
root@kali:~/rhion# mkdir steg

root@kali:~/rhion# cp recup_dir.1/f0104057.jpg steg/

root@kali:~/rhion# cp recup_dir.1/f0104249.jpg steg/

root@kali:~/rhion# cp recup_dir.1/f0105065.jpg steg/

root@kali:~/rhion# cp recup_dir.1/f0335081.jpg steg/

root@kali:~/rhion# cd recup_dir.1

root@kali:~/rhion/recup_dir.1# md5sum f0105065.jpg
6bd0e9bd4fb4a738f9ca4c351a853281  f0105065.jpg
```

```
root@kali:~/rhion/steg# apt-get install exiftool
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
Note, selecting 'libimage-exiftool-perl' instead of 'exiftool'
libimage-exiftool-perl is already the newest version (13.55+dfsg-1).
libimage-exiftool-perl set to manually installed.
Solving dependencies ... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
```

```
root@kali:~/rhion/steg# exiftool f0104057.jpg
ExifTool Version Number         : 13.55
File Name                       : f0104057.jpg
Directory                       : .
File Size                       : 96 kB
File Modification Date/Time     : 2026:10:26 15:34:57+01:00
File Access Date/Time           : 2026:10:26 15:34:57+01:00
File Inode Change Date/Time     : 2026:10:26 15:34:57+01:00
File Permissions                : -rw-r--r--
File Type                       : JPEG
File Type Extension             : jpg
MIME Type                       : image/jpeg
JFIF Version                    : 1.01
Resolution Unit                 : None
X Resolution                    : 0
Y Resolution                    : 0
Comment                         : copyright 2000 philg@mit.edu
Image Width                     : 528
Image Height                    : 792
Encoding Process                : Baseline DCT, Huffman coding
Bits Per Sample                 : 8
Color Components                : 3
Y Cb Cr Sub Sampling            : YCbCr4:4:4 (1 1)
Image Size                      : 528×792
Megapixels                      : 0.418
```

Metasploitable 2
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```
root@kali:~/rhion/steg/stegdetect# git clone https://github.com/poizan42/stegdetect
Cloning into 'stegdetect'...
remote: Enumerating objects: 419, done.
remote: Total 419 (delta 0), reused 0 (delta 0), pack-reused 419 (from 1)
Receiving objects: 100% (419/419), 1.38 MiB | 23.00 KiB/s, done.
Resolving deltas: 100% (59/59), done.

root@kali:~/rhion/steg/stegdetect# autoreconf -f -i
autoreconf: warning: autoconf input should be named 'configure.ac', not 'configure.in'
aclocal: warning: autoconf input should be named 'configure.ac', not 'configure.in'
aclocal: warning: autoconf input should be named 'configure.ac', not 'configure.in'
acinclude.m4:2: warning: underquoted definition of AC_C_LONG_LONG
acinclude.m4:2:   run info Automake 'Extending aclocal'
acinclude.m4:2:   or see https://www.gnu.org/software/automake/manual/automake.html#Extending-aclocal
configure.in:3: warning: AM_INIT_AUTOMAKE: two- and three-arguments forms are deprecated.
./lib/autoconf/general.m4:2490: AC_DIAGNOSE is expanded from...
aclocal.m4:429: AM_INIT_AUTOMAKE is expanded from...
configure.in:3: the top level
configure.in:4: warning: 'AM_CONFIG_HEADER': this macro is obsolete.
configure.in:4: You should use the 'AC_CONFIG_HEADERS' macro instead.
aclocal.m4:708: AM_CONFIG_HEADER is expanded from...
configure.in:4: the top level
configure.in:57: warning: The macro 'AC_HEADER_STDC' is obsolete.
configure.in:57: You should run autoupdate.
./lib/autoconf/headers.m4:664: AC_HEADER_STDC is expanded from...
configure.in:57: the top level
configure.in:68: warning: The macro 'AC_STRUCT_ST_RDEV' is obsolete.
configure.in:68: You should run autoupdate.
./lib/autoconf/types.m4:1058: AC_STRUCT_ST_RDEV is expanded from...
./lib/autoconf/types.m4:1058: AC_STRUCT_ST_RDEV is expanded from...
configure.in:68: the top level
configure.in:69: warning: The macro 'AC_TRY_COMPILE' is obsolete.
configure.in:69: You should run autoupdate.
./lib/autoconf/general.m4:2903: AC_TRY_COMPILE is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...
lib/m4sugar/m4sh.m4:696: AS_IF is expanded from...
./lib/autoconf/general.m4:2305: AC_CACHE_VAL is expanded from...
./lib/autoconf/general.m4:2326: AC_CACHE_CHECK is expanded from...
acinclude.m4:26: AC_STRUCT_TIMEZONE_DAYLIGHT is expanded from...
configure.in:69: the top level
configure.in:69: warning: The macro 'AC_TRY_LINK' is obsolete.
configure.in:69: You should run autoupdate.
./lib/autoconf/general.m4:2976: AC_TRY_LINK is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...
lib/m4sugar/m4sh.m4:696: AS_IF is expanded from...
./lib/autoconf/general.m4:2305: AC_CACHE_VAL is expanded from...
./lib/autoconf/general.m4:2326: AC_CACHE_CHECK is expanded from...
acinclude.m4:26: AC_STRUCT_TIMEZONE_DAYLIGHT is expanded from...
configure.in:69: the top level
configure.in:76: warning: The macro 'AC_TRY_RUN' is obsolete.
configure.in:76: You should run autoupdate.
./lib/autoconf/general.m4:3053: AC_TRY_RUN is expanded from...
lib/m4sugar/m4sh.m4:689: _AS_IF_ELSE is expanded from...
lib/m4sugar/m4sh.m4:696: AS_IF is expanded from...
configure.in:153: warning: AC_OUTPUT should be used without arguments.
configure.in:153: You should run autoupdate.
autoheader: warning: autoconf input should be named 'configure.ac', not 'configure.in'
autoheader: warning: WARNING: Using auxiliary files such as 'acconfig.h', 'config.h.bot'
autoheader: WARNING: and 'config.h.top', to define templates for 'config.h.in'
autoheader: WARNING: is deprecated and discouraged.
autoheader:
autoheader: WARNING: Using the third argument of 'AC_DEFINE_UNQUOTED' and
autoheader: WARNING: 'AC_DEFINE' allows one to define a template without
autoheader: WARNING: 'acconfig.h':
autoheader:
autoheader: WARNING:   AC_DEFINE([NEED_FUNC_MAIN], 1,
autoheader:              [Define if a function 'main' is needed.])
autoheader:
autoheader: WARNING: More sophisticated templates can also be produced, see the
autoheader: WARNING: documentation.
automake: warning: autoconf input should be named 'configure.ac', not 'configure.in'
configure.in:6: warning: AM_INIT_AUTOMAKE: two- and three-arguments forms are deprecated.  For more info, see:
configure.in:6: https://www.gnu.org/software/automake/manual/automake.html#Modernize-AM_005fINIT_005fAUTOMAKE-invocation
configure.in:15: installing './compile'
Makefile.am:30: warning: 'INCLUDES' is the old name for 'AM_CPPFLAGS' (or '*_CPPFLAGS')
automake: warning: autoconf input should be named 'configure.ac', not 'configure.in'

root@kali:~/rhion/steg/stegdetect# linux32 ./configure
checking build system type... i686-pc-linux-gnu
checking host system type... i686-pc-linux-gnu
checking target system type... i686-pc-linux-gnu
checking for a BSD-compatible install... /usr/bin/install -c
```

```
checking whether sleep supports fractional seconds ... yes
checking filesystem timestamp resolution ... 0.01
checking whether build environment is sane ... yes
checking for a race-free mkdir -p ... /usr/bin/mkdir -p
checking for gawk ... gawk
checking whether make sets $(MAKE) ... yes
checking whether make supports nested variables ... yes
checking xargs -n works ... yes
checking whether UID '0' is supported by ustar format ... yes
checking whether GID '0' is supported by ustar format ... yes
checking how to create a ustar tar archive ... gnutar
checking whether make supports the include directive ... yes (GNU style)
checking for gcc ... gcc
checking whether the C compiler works ... yes
checking for C compiler default output file name ... a.out
checking for suffix of executables ...
checking whether we are cross compiling ... no
checking for suffix of object files ... o
checking whether the compiler supports GNU C ... yes
checking whether gcc accepts -g ... yes
checking for gcc option to enable C23 features ... none needed
checking whether gcc understands -c and -o together ... yes
checking dependency style of gcc ... none
checking dependency style of gcc ... none
checking for g++ ... g++
checking whether the compiler supports GNU C++ ... yes
checking whether g++ accepts -g ... yes
checking whether byte ordering is bigendian ... no
checking that generated files are newer than configure ... done
configure: creating ./config.status
config.status: creating Makefile
config.status: creating config.h
config.status: executing depfiles commands
=== configuring in jpeg-6b (/root/rhion/steg/stegdetect/jpeg-6b)
configure: running /bin/bash ./configure --disable-option-checking '--prefix=/usr/local'  --cache-file=/dev/null --srcdir=.
checking for gcc ... gcc
checking whether the C compiler (gcc  ) works ... no
configure: error: installation or configuration problem: C compiler cannot create executables.
configure: error: ./configure failed for jpeg-6b

root@kali:~/rhion/steg/stegdetect# linux32 make
make  all-recursive
make[1]: Entering directory '/root/rhion/steg/stegdetect'
Making all in jpeg-6b
make[2]: Entering directory '/root/rhion/steg/stegdetect/jpeg-6b'
make[2]: *** No rule to make target 'all'.  Stop.
make[2]: Leaving directory '/root/rhion/steg/stegdetect/jpeg-6b'
make[1]: *** [Makefile:638: all-recursive] Error 1
make[1]: Leaving directory '/root/rhion/steg/stegdetect'
make: *** [Makefile:455: all] Error 2
```